Mock Olympiad (IMC), Round 1

June 28, 2026
Time allowed: 5 hours. Each problem is worth 10 points.

Problem 1. Let z be a complex number with $|z+1|>2$. Prove that $|z^3+1|>1$.

Problem 2. Let n be a fixed positive integer. Determine the smallest possible rank of an $n \times n$ matrix that has zeros along the main diagonal and strictly positive real numbers off the main diagonal.

Problem 3. There are $2n$ students in a school ($n \in \mathbb{N}$, $n \geq 2$). Each week n students go on a trip. After several trips the following condition was fulfilled: every two students were together on at least one trip. What is the minimum number of trips needed for this to happen?

Problem 4. Let $f : \mathbb{R} \to \mathbb{R}$ be a continuously differentiable function that satisfies $f'(t) > f(f(t))$ for all $t \in \mathbb{R}$. Prove that $f(f(f(t))) \leq 0$ for all $t \geq 0$.

Problem 5. Let a be a rational number and let n be a positive integer. Prove that the polynomial $X^{2^n}(X+a)^{2^n}+1$ is irreducible in the ring $\mathbb{Q}[X]$ of polynomials with rational coefficients.